

4 June 2024

14h00

DJ EXAMS HALL

EXAMS OFFICE USE ONLY

University of the Witwatersrand, Johannesburg

Course or topic No(s)

COMS3003A

Course or topic name(s)
Paper Number & Title

Formal Languages and Automata

Examination/to be held during month(s) of

June 2024

Year of Study(Art & Science leave blank)

3rd

Degrees/Diplomas for which this course is
prescribed (Bsc (Eng.) should indicate which
branch)Faculty/ies presenting candidates

BSc Computer Science

Science

Internal examiner(s) and telephone extension
number(s)Dmitry Shkatov
76178

External examiner(s)

Prof. Thomas Meyer (University of Cape Town)

Special materials required
(graph/music/drawing paper maps. diagrams.
tables. computer cards, etc)

NIL

Time allowance

2 hours

This is a **closed book** examination.

The exam is out of 120

You may not use a calculator in the exam.

If you have any questions about the exam, feel
free to ask.

- As Internal Examiner/Head of Department, I certify that this question paper is in final form, as approved by the External Examiner, and is for reproduction.
- As Internal Examiner/Head of Department, I certify that this question paper is in final form and is ready for reproduction.
- application to formal examinations as approved by an external examiner.

COMS 3003A

Final exam

4 June, 2024

Time allowance: 2 hours
Total number of points: 120

1. (20 points)

- (a) (5 points) Are there any types of data that can be encoded in some finite alphabet, but not in the binary alphabet $\{0, 1\}$?
- (b) (5 points) Give the formal definition of a decision problem. Assume that we use the binary alphabet for encoding data.
- (c) (5 points) What does it mean that a Turing machine M solves a decision problem P ?
- (d) (5 points) What distinguishes a decider from an arbitrary Turing machine?

2. (10 points) Design a Turing machine that checks if the input binary string is a palindrome. (A palindrome is a word that reads the same forwards and backwards; e.g., 0110 is a palindrome, but 0111 is not; the empty word is a palindrome).

3. (20 points) When answering this question, you may assume Cantor's theorem, which says that, for every set X , the cardinality of the power set of X is strictly greater than the cardinality of X . You may also assume that the set of finite strings in a finite alphabet is countable.

- (15 points) Prove that the cardinality of the set of all decision problems is strictly greater than the cardinality of the set of all deciders.
- (5 points) Explain why this implies the existence of problems unsolvable by Turing machines.

$$|P| > |D|$$

4. (20 points) In any part of this question, if you define a Turing machine and claim that it solves a particular problem, you have to prove your claim.

(a) (10 points) Prove, using diagonalization, that the self-rejection problem for Turing machines is algorithmically unsolvable, i.e., that the language $SR = \{\langle M \rangle \mid M(\langle M \rangle) = 0\}$ is undecidable. You may not assume unsolvability of any other problem.

(b) (10 points) Prove, using unsolvability of the problem from (a), that the problem of determining whether a Turing machine rejects an input word is unsolvable, i.e., that the language $R = \{\langle M, w \rangle \mid M(w) = 0\}$ is undecidable. If you define a Turing machine and claim that it solves a particular problem, you have to prove your claim.

5. (10 points) Prove that the problem of determining whether a Turing machine ever writes a 0 is algorithmically unsolvable. You may assume that the following problems are unsolvable:

- self-acceptance for Turing machines;
- acceptance for Turing machines.

If you define a Turing machine and claim that it solves a particular problem, you have to prove your claim.

6. (10 points) Consider the formula $\varphi = \forall x \forall y \forall z (R(x, y) \wedge R(y, z) \rightarrow R(x, z)) \wedge \forall x \exists y R(x, y) \wedge \forall x \neg R(x, x)$. Decide if φ is true in the following models:

- (5 points) $M = (D, I)$ where $D = \mathbb{N}$ (the set of natural numbers) and $I(R) = \{\langle n, m \rangle : n < m\}$;
- (5 points) $M = (D, I)$ where $D = \{a\}$ and $I(R) = \emptyset$.

7. (10 points) Prove that the formula $\exists x P(x) \wedge \exists x Q(x) \rightarrow \exists x (P(x) \wedge Q(x))$ is not valid.

8. (20 points) In the proof of undecidability of the first-order logic, we describe computations of Turing machines on the empty string ϵ . When describing a Turing machine $M = (Q, \Sigma, \Gamma, q_0, q_1, q_2, \delta)$, we use the following vocabulary (remember that every first-order language contains the equality symbol $=$):

- the constant symbol 0 ('zero');
- the unary function symbol S ('successor' on natural numbers);
- for every $k \in \{0, \dots, |Q|\}$, a unary predicate letter $Q_k(x)$ ('at step x , the machine M is in state q_k ');
- a binary predicate letter $C(x, y)$ ('at step x , the machine M scans cell y ');
- for every $k \in \{0, \dots, |\Gamma|\}$, a binary predicate letter $S_k(x, y)$ ('at step x , cell y contains symbol s_k ').

When describing computations of Turing machines, we will be making the following assumptions:

- the tape is one-way infinite (hence the cells can be numbered 0, 1, 2, etc.);
- machines can move left and right (no other types of head movement are allowed);
- the left-most cell, i.e., cell 0, always contains an end-of-tape marker, which is a special tape symbol;
- states are always called q_0, q_1, q_2 , etc.;
- the starting state is q_0 , the accepting state is q_1 , and the rejecting state is q_2 ;
- the tape symbols are always called s_0, s_1, s_2 , etc.
- s_0 is the blank;
- s_1 is the end-of-tape marker.

Let $M = (Q, \Sigma, \Gamma, q_0, q_1, q_2, \delta)$ be a Turing machine. Write formulas saying the following, in the vocabulary listed above:

- (5 points) A formula saying that M halts.
- (5 points) A formula describing the initial configuration of M on the empty word ϵ .
- (10 points) Suppose that $q_i s_k \rightarrow q_j s_l R$ is an instruction from δ . Write a formula describing the effect of executing this instruction.